Chemical formulas – what a combo! 10a

Note: You will need to refer to a copy of the periodic table for this worksheet.

When you write the formula for a compound, you need to balance its charges by writing numbers as subscripts. For example, Li^+ has a one plus charge and O^{2-} has a two minus charge. When you write a little '2' after the lithium symbol, you are saying that there are two lithiums now. Since each one has a plus one charge, two of them have a two plus charge. And that balances exactly the two minus charge from O^{2-}. Therefore, the formula combining Li^+ and O^{2-} is Li_2O, lithium oxide.

$Li^+ O^{2-} \longrightarrow Li^{1+} O^{2-} \longrightarrow Li^{1+}{}_2O^{2-} \longrightarrow Li_2O_1 \longrightarrow Li_2O$

1 Look at the table about chemicals below. Write down each combination in words. Then write down the chemical formula. Some have been done for you already.

	Chloride Cl^-	Oxide O^{2-}	Iodide I^-	Sulfide S^{2-}	Nitrate NO_3^-	Sulfate SO_4^{2-}	Hydroxide OH^-	Carbonate CO_3^{2-}
Lithium Li^+	Lithium chloride LiCl	Lithium oxide Li_2O	Lithium iodide LiI	Lithium sulfide Li_2S	Lithium nitrate $LiNO_3$	Lithium sulfate Li_2SO_4	Lithium hydroxide LiOH	Lithium carbonate Li_2CO_3
Hydrogen H^+			HI			H_2SO_4		
Magnesium Mg^{2+}			MgI_2		$Mg(NO_3)_2$			
Sodium Na^+		Na_2O			$NaNO_3$			
Copper Cu^{2+}				CuS				$CuCO_3$
Iron Fe^{3+}	$FeCl_3$	Fe_2O_3						
Calcium C^{2+}				CaS			$Ca(OH)_2$	
Zinc Zn^+	$ZnCl_2$					$ZnSO_4$		
Aluminium Al^{3+}			AlI_3					$Al_2(CO_3)_3$

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Chemical formulas – what a combo! 10b

2 Iron compounds can have two different formulas. For example, another formula for iron chloride is $FeCl_2$. This is often called iron II chloride in order to distinguish it from the iron III chloride shown in the table. Write down the formulas of the other iron II compounds from the table.

_____

3 Copper compounds can also have two formulas: copper I compounds and copper II compounds.

a State whether copper I or copper II compounds are in the table. _____

b Write the formulas for the other copper compounds.

_____

4 Use the table above and your periodic table to write the formula for each of these compounds:

a lead nitrate _____

b potassium hydroxide _____

c silver bromide _____

d barium chloride _____

e potassium iodide _____

5 Name the following compounds:

a $AgNO_3$ _____

b K_2S _____

c BaS _____

d AgI _____

e KI _____

6 Find the following chemical names in the puzzle below. The 13 remaining letters will spell out an important chemical list.

lithium Li hydrogen magnesium Mg sodium Na copper Cu iron Fe calcium
zinc Zn aluminium chloride oxide iodide sulfide nitrate sulfate hydroxide
carbonate lead potassium silver Ag barium bromide

Answer: _____

C	H	L	O	R	I	D	E	Z	S	O	D	I	U	M
A	E	T	A	F	L	U	S	I	M	X	A	R	B	A
L	I	T	H	I	U	M	P	N	G	I	E	O	R	G
C	O	E	Y	R	L	I	D	C	I	D	L	N	O	N
I	D	E	D	I	F	L	U	S	F	E	O	D	M	E
U	I	T	R	A	L	U	M	I	N	I	U	M	I	S
M	D	C	O	P	P	E	R	E	V	L	I	S	D	I
I	E	A	G	M	U	I	S	S	A	T	O	P	E	U
C	U	A	E	H	Y	D	R	O	X	I	D	E	C	M
L	B	G	N	I	T	R	A	T	E	E	N	A	Z	N
B	A	R	I	U	M	E	T	A	N	O	B	R	A	C

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